

27-4-26

# \* Producer - Consumer Problem:

```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define BUF_SIZE 5

int buffer[BUF_SIZE];
int in = 0, out = 0;
sem_t empty, full;
pthread_mutex_t mutex;

void *producer(void *arg) {
    int item;
    for(int i = 0; i < 10; i++) {
        item = rand() % 100;
        sem_wait(&empty);
        pthread_mutex_lock(&mutex);
        buffer[in] = item;
        printf("Produced %d at %d\n", item, in);
        in = (in + 1) % BUF_SIZE;
        pthread_mutex_unlock(&mutex);
        sem_post(&full);
        sleep(1);
    }
    return NULL;
}
```

---



```

void *consumer (void *arg){
    int item;
    for(int i=0; i<10; i++){
        sem_wait (& full);
        pthread_mutex_lock (& mutex);
        item = buffer[out];
        printf("Consumed %d from %d\n", item,
            out);
        out = (out + 1) % BUF_SIZE;
        pthread_mutex_unlock (& mutex);
        sem_post (& empty);
        sleep(2);
    }
    return NULL;
}

```

```

int main(){
    pthread_t prod, cons;
    sem_init (& empty, 0, BUF_SIZE);
    sem_init (& full, 0, 0);
    pthread_mutex_init (& mutex, NULL);
    pthread_create (& prod, NULL, producer, NULL);
    pthread_create (& cons, NULL, consumer, NULL);
    pthread_join (prod, NULL);
    pthread_join (cons, NULL);
    sem_destroy (& empty);
    sem_destroy (& full);
    pthread_mutex_destroy (& mutex);
    return 0;
}

```



O/P :  
Produced 41 at 0  
Consumed 41 from 0  
Produced 67 at 1  
Consumed 67 from 1  
Produced 34 at 2  
Produced 0 at 3  
Consumed 34 from 2  
Produced 69 at 4  
Consumed 0 from 3  
Produced 78 at 1  
Produced 58 at 2  
Consumed 69 from 4  
Produced 62 at 3  
Produced 64 at 4  
Consumed 24 from 0  
Consumed 78 from 1  
Consumed 58 from 2  
Consumed 62 from 3  
Consumed 64 from 4.

---

Per  
27/4/24